

★ WEEK 11 · PART 2 of 3

The Ornstein-Uhlenbeck Process

Mean Reversion, in Continuous Time

The spring mechanism · κ , θ , σ · half-life · a continuous AR(1)

Brownian motion wanders without an anchor. The Ornstein-Uhlenbeck process adds a drift that pulls the state back toward a mean — the continuous-time model of a mean-reverting spread.

2

PART 2 · THE OU PROCESS

A Spring That Pulls to the Mean

The OU stochastic differential equation, the reversion mechanism, the roles of its three parameters, the half-life, and the exact discretization that makes OU a continuous-time AR(1).

From Wandering to Reverting

Part 1 built SDEs; Part 2 adds a pull

1

We have the SDE template

Drift plus diffusion. Brownian motion and GBM both wander — nothing pulls them back to a level.

2

But spreads revert

A cointegrated pairs spread doesn't drift away; it is tethered to a mean. We need an SDE that captures that pull.

3

The OU process is that model

An SDE whose drift points back toward a mean — continuous-time mean reversion, and the continuous cousin of the AR(1).

Part 2 in one line: add a mean-reverting drift and Brownian motion becomes the OU process.

The Ornstein-Uhlenbeck Process

An SDE with a pull toward the mean

$$\text{OU: } dX_t = \underbrace{\kappa(\theta - X_t)}_{\text{pull to the mean}} dt + \sigma dW_t$$

The drift

- It is proportional to the gap between the mean and X .
- When X is above the mean, the drift points down.
- When X is below, it points up — always toward the mean.

The diffusion

- A constant-size Brownian shock, as before.
- It pushes X around at random.
- The drift fights it, pulling X back home.

From GBM to OU

One wanders, the other is tethered

GBM (Part 1)

- The drift pushes in one direction, forever.
- The process wanders with no anchor.
- Good for prices, which trend and compound.

OU (this part)

- The drift reverses sign across the mean.
- The process is pulled back whenever it strays.
- Good for spreads, which revert to an equilibrium.

The Reversion Mechanism

A spring pulling toward the mean

Think of the state attached to the mean by a spring. Stretch it away and the spring pulls back — harder the farther you stretch. That restoring force is exactly the OU drift, κ times the gap.

The spring analogy

- The mean is the spring's resting position.
- Displacement creates a restoring force toward it.
- Bigger displacement, stronger pull — proportional.

In the SDE

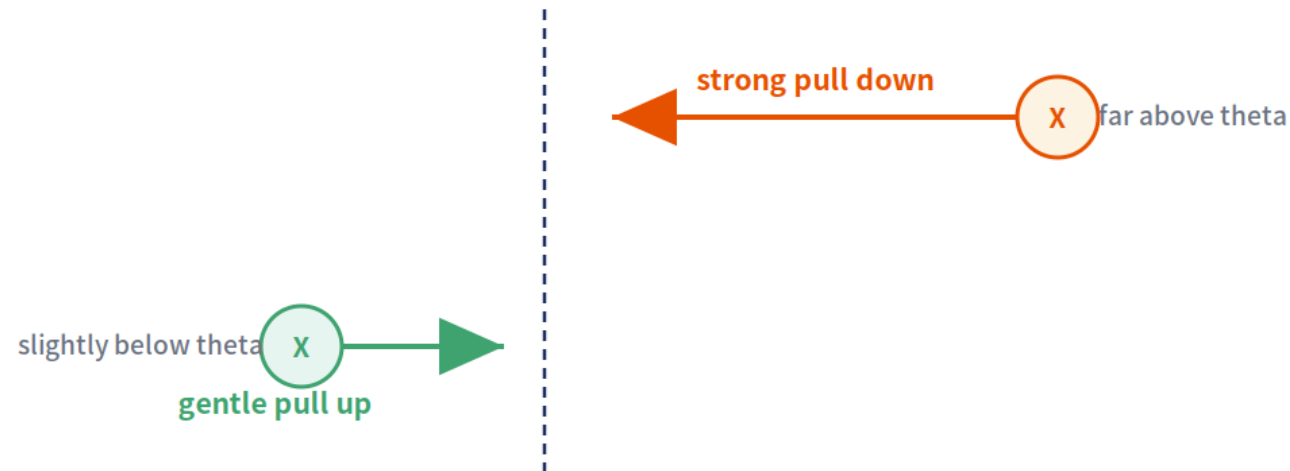
- The restoring force is the drift, κ times (mean minus X).
- The noise keeps knocking X away from the mean.
- The two balance into fluctuations around the mean.

The Reversion Mechanism: a Spring

The drift always points toward the mean and grows with distance. Far above, a strong pull down; slightly below, a gentle pull up. The gap sets direction; the speed kappa sets strength.

The Reversion Mechanism: a Spring

The drift always points back toward theta, and pulls harder the farther X strays.



$$\text{drift} = \text{kappa} (\text{theta} - X)$$

The gap (theta - X) sets the direction; kappa sets the strength. Bigger gap, stronger restoring force - exactly like a spring. The noise sigma dW pushes X around, but the spring keeps dragging it home.

The Three OU Parameters

What kappa, theta, and sigma control

Parameter	Role	What it controls
kappa	reversion speed	how fast X is pulled back to the mean
theta	long-run mean	the level X reverts toward
sigma	volatility	the size of the random shocks

Three numbers fully describe an OU process: where it settles, how strongly, and how noisily.

What kappa Controls

The speed of the pull

Large kappa

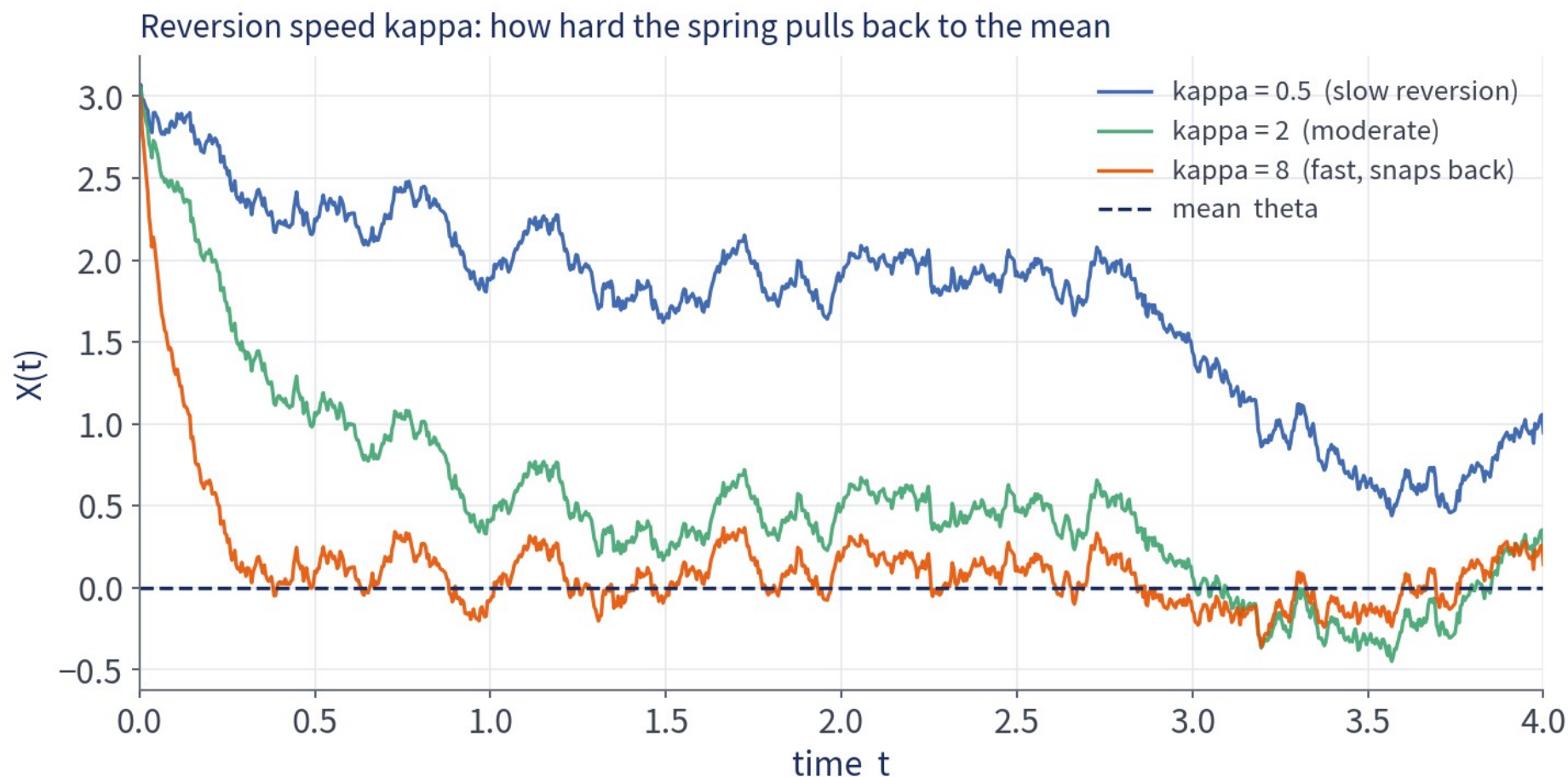
- A strong, fast spring.
- X snaps back to the mean quickly.
- Short excursions, short half-life.

Small kappa

- A weak, slow spring.
- X drifts far and lingers before returning.
- Long excursions, long half-life.

Reversion Speed in Pictures

Three OU paths starting away from the mean, differing only in kappa. A large kappa snaps back fast; a small one wanders and returns slowly. The mean and noise are identical.



The Stationary Distribution

Where OU settles over time

$$\text{stationary law: } X_{\infty} \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{2\kappa}\right)$$

The long-run law

- Left alone, OU settles into a Gaussian distribution.
- Its mean is theta — the reversion level.
- Its variance balances noise against reversion speed.

Reading the variance

- More noise (sigma) widens the spread.
- Faster reversion (kappa) tightens it.
- The spread is sigma-squared over twice kappa.

OU Half-Life

How long to close half the gap

$$\text{half-life: } t_{1/2} = \frac{\ln 2}{\kappa} \quad (\text{time to close half the gap})$$

What it is

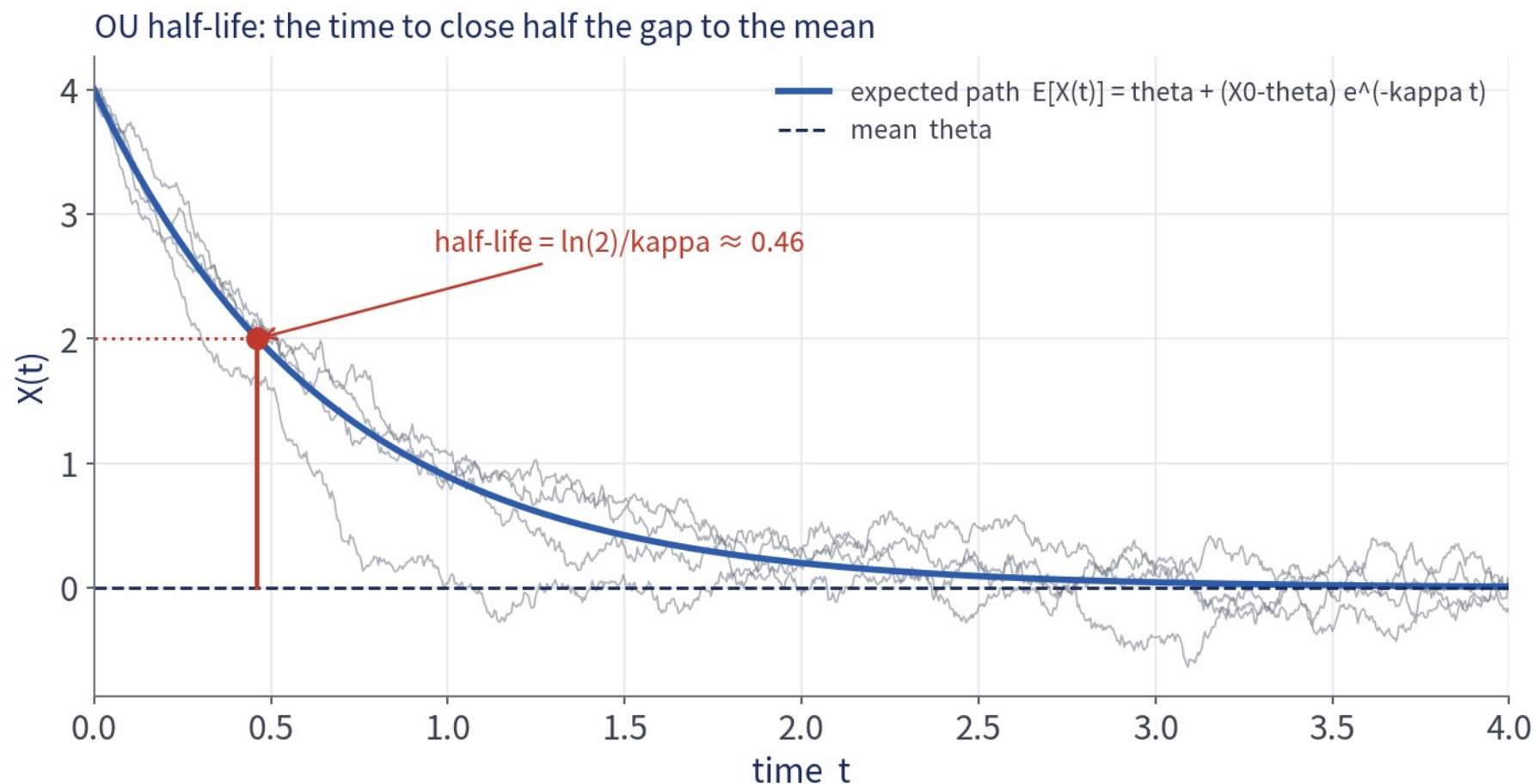
- The time for the expected gap to the mean to halve.
- It is the natural of the reversion speed.
- Short half-life means quick mean reversion.

Why traders care

- It sets the horizon of a mean-reversion trade.
- It tells you how long to expect to hold.
- We first met it as a discrete idea in the pairs thread.

The Half-Life, Illustrated

Starting away from the mean, the expected path decays exponentially back toward it. The half-life marks where half the initial gap has closed — the natural clock of the process.



The OU Process in Action

Pulled back, unlike Brownian motion

The defining behavior: OU stays near its mean. Shocks push it away, but the spring always drags it back. Over time it fluctuates in a band around the mean, rather than wandering off like Brownian motion.

What you see

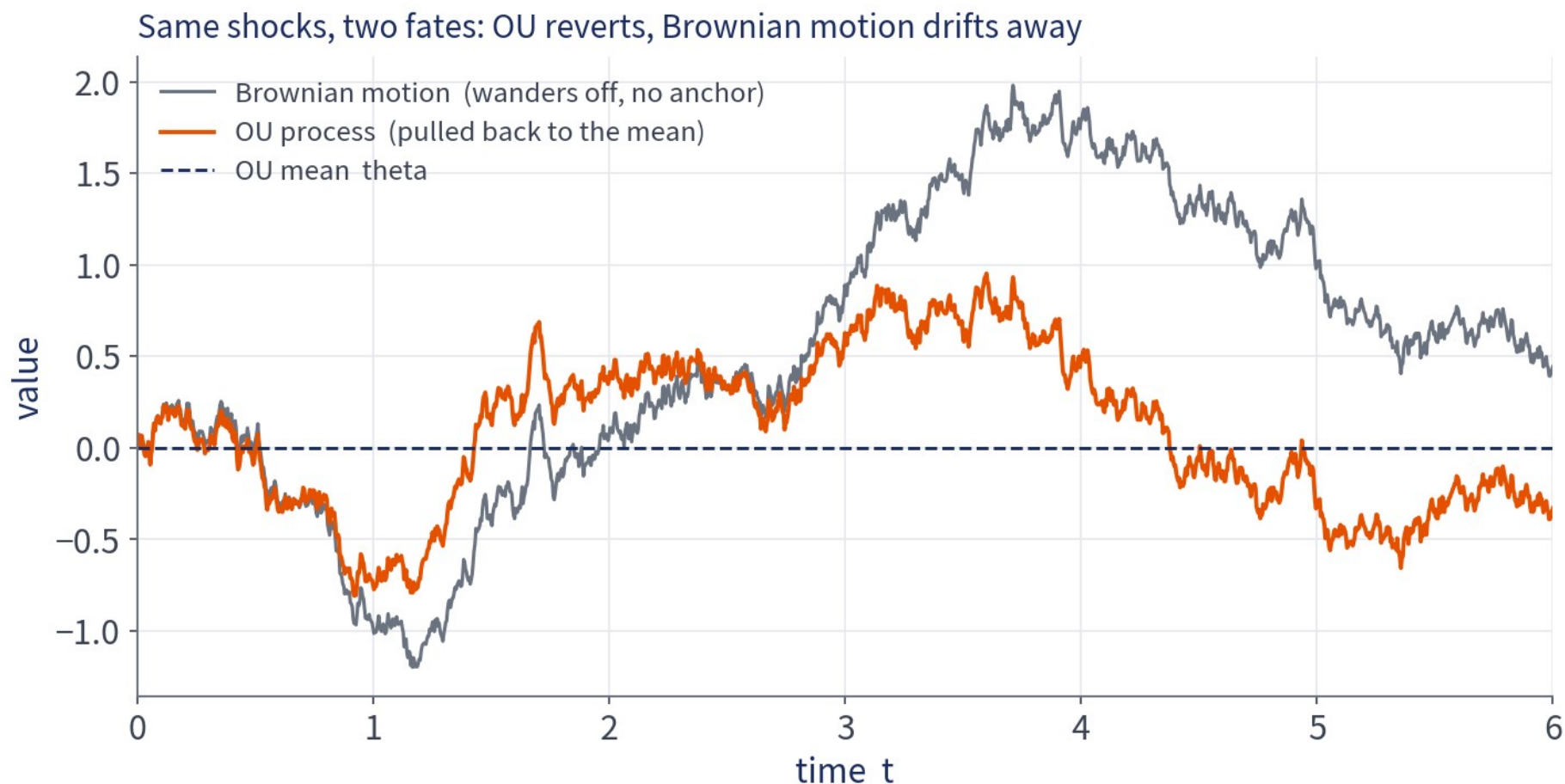
- A path that hovers around a central level.
- Excursions that are pulled back, not sustained.
- A stable, bounded spread of values.

Why it fits spreads

- A cointegrated spread behaves exactly this way.
- It is tethered to an equilibrium, not free to drift.
- OU gives a clean model of that tether.

OU vs Brownian Motion

Driven by the very same shocks, the OU process (orange) is pulled back to its mean while Brownian motion (grey) wanders off. The difference is entirely the mean-reverting drift.



Discretization: OU Is a Continuous AR(1)

The bridge from this week back to Part I

Sample the OU process at fixed intervals and you get exactly an AR(1) — the very model from the start of the course. Continuous-time mean reversion and discrete-time autoregression are the same idea.

The correspondence

- Observing OU on a grid gives an AR(1) recursion.
- The persistence ϕ is set by the reversion speed.
- The mean and noise carry over directly.

Why it is satisfying

- Two whole frameworks turn out to describe one process.
- The continuous view gives intuition (a spring).
- The discrete view gives an easy estimation route.

The Exact Discretization

OU sampled on a grid

$$X_{t+\Delta t} = \theta + e^{-\kappa\Delta t} (X_t - \theta) + \varepsilon_t \quad \varepsilon_t \sim \mathcal{N}\left(0, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta t})\right)$$

Reading it

- Next value = mean + a decayed gap + Gaussian noise.
- The decay factor is the exponential of minus kappa times dt.
- This is exactly the AR(1) form, with a mean.

It is exact, not approximate

- No Euler error — the OU SDE solves in closed form.
- The formula holds for any step size dt.
- So you can estimate it cleanly from sampled data.

OU Is a Continuous-Time AR(1)

Discretize the OU SDE and the AR(1) from Part I reappears: the persistence ϕ equals the exponential of minus κ times dt . Estimate the AR(1), then read off the continuous parameters.

OU Is a Continuous-Time AR(1)

Discretize the OU SDE and you recover the AR(1) from Part I - the bridge back to discrete time.

Continuous time (OU)

$$dX = \kappa(\theta - X) dt + \sigma dW$$

κ - reversion speed

θ - long-run mean

σ - volatility

$$\text{half-life} = \ln(2) / \kappa$$

discretize

step size dt

Discrete time (AR(1))

$$X_{t+1} = \theta + \phi(X_t - \theta) + \epsilon$$

$\phi = e^{(-\kappa dt)}$ - persistence

same θ - the mean

ϵ - Gaussian innovation

$$\kappa = -\ln(\phi) / dt$$

Estimate the AR(1) in discrete time (easy), then read off the continuous OU parameters - reversion speed and half-life - from ϕ . Two languages for one mean-reverting process.

The Parameter Mapping

From AR(1) persistence to reversion speed

$$\text{it is an AR(1): } \phi = e^{-\kappa \Delta t} \iff \kappa = -\frac{\ln \phi}{\Delta t}$$

One key relation

- Persistence ϕ is the exponential of minus κ times Δt .
- So κ is minus the log of ϕ , divided by Δt .
- High persistence (ϕ near 1) means slow reversion.

What it lets you do

- Fit an AR(1) — a one-line regression.
- Convert ϕ into κ and a half-life.
- Report continuous-time quantities from discrete data.

Why This Bridge Matters

Estimate discretely, interpret continuously

The practical route

- Real data arrive on a grid — daily, hourly.
- AR(1) estimation on that grid is trivial.
- The mapping then hands you the OU parameters.

The conceptual gain

- The continuous model gives the spring intuition and half-life.
- The discrete model gives a clean estimator.
- You get the best of both descriptions.

What Part 3 Adds

From model to trade

Pairs Stage V

- Model the cointegrated spread as an OU process.
- Read off its mean, half-life, and typical range.
- Turn those into principled entry and exit rules.

Estimation and cautions

- How to fit OU parameters from a spread series.
- Why fitting is easy but trusting is hard.
- The bridge to interest-rate models (Week 12).

Simulate and Estimate an OU Process

Direct Claude Code to build OU intuition

GOAL Simulate an OU process, then recover its parameters via the AR(1) bridge — and check the half-life.

[INPUT] Prompt · Claude Code

Write and run `ou_estimate.py` that:

1. simulates an OU process for known κ , θ , σ (exact discretization);
2. estimates it by regressing X_{t+1} on X_t (an AR(1)) and recovering κ , θ , σ ;
3. converts the AR(1) ϕ into κ via $\kappa = -\ln(\phi)/dt$, and the half-life;
4. plots the path with the mean and the half-life horizon marked.

Confirm the recovered parameters match the truth.

How to work with it

- ϕ near 1 means slow reversion.
- Recovered κ should match the input.
- Half-life = $\ln(2)/\kappa$.
- Part 3 applies this to a real spread.

Deliverable: `ou_estimate.py` + recovered κ , θ , σ , and a half-life estimate.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 What does the OU drift do when X is above the mean?

Pulls it down

Q2 What is the OU half-life in terms of the reversion speed?

$\ln(2) / \kappa$

Q3 Sampled on a grid, an OU process is exactly what model?

An AR(1)

All three come from the reversion drift, the half-life formula, and the discretization.

Three Misconceptions to Avoid

Where OU intuition slips

Common myths

- "OU and Brownian motion look basically the same."
- "A larger sigma means slower reversion."
- "Discretizing OU is only an approximation."

The reality

- OU is tethered to a mean; Brownian motion wanders off.
- Reversion speed is κ ; sigma only sets the noise size.
- OU's discretization is EXACT — a true AR(1), any Δt .

OU at a Glance

Everything on one card

Quantity	Formula / form	Meaning
Dynamics	$dX = \kappa(\theta - X)dt + \sigma dW$	the OU SDE
Long-run mean	θ	where it settles
Long-run variance	$\sigma^2 / (2\kappa)$	spread around the mean
Half-life	$\ln(2) / \kappa$	time to close half the gap
Discrete form	AR(1), $\phi = e^{(-\kappa dt)}$	the estimation bridge

Keep this card: it is the whole of the OU process in five lines.

Three Things to Carry Forward

Part 2 in one screen

1

OU is Brownian motion with a spring

Its drift, κ times the gap to the mean, pulls the state back toward θ whenever it strays. The noise pushes; the spring pulls; the two balance into fluctuations around the mean.

2

Three parameters, one half-life

Reversion speed κ , mean θ , and volatility σ fully describe it. The half-life, $\log_2 2 / \kappa$, is the natural clock of the reversion — and the horizon of a trade.

3

OU is a continuous-time AR(1)

Sampled on a grid it is exactly an AR(1) with persistence $e^{-\kappa \Delta t}$. Estimate the AR(1), then read off the continuous parameters — the bridge back to Part I.

Next — Part 3: model the pairs spread as an OU process (Stage V), estimate it, and trade it.